Models 3–4: Linear Regression and Random Forest — Methods and Performance

Methods

- **Models:** Ordinary least squares (OLS) and tuned Random Forest (RF)
- **Predictor sets:** 15 covariates vs. 15 covariates + 12 behaviors
- **Split:** 80/20 train/test (2,235 / 559), outcome-stratified
- **Preprocessing:** Median imputation; OLS predictors standardized
- **Validation:** Repeated 5-fold CV (25 folds); nested tuning for RF
- **Tuning:** 12 RF configurations, RMSE criterion, 3-fold inner CV
- **Added value:** 2,000 paired test-set bootstrap resamples

Performance

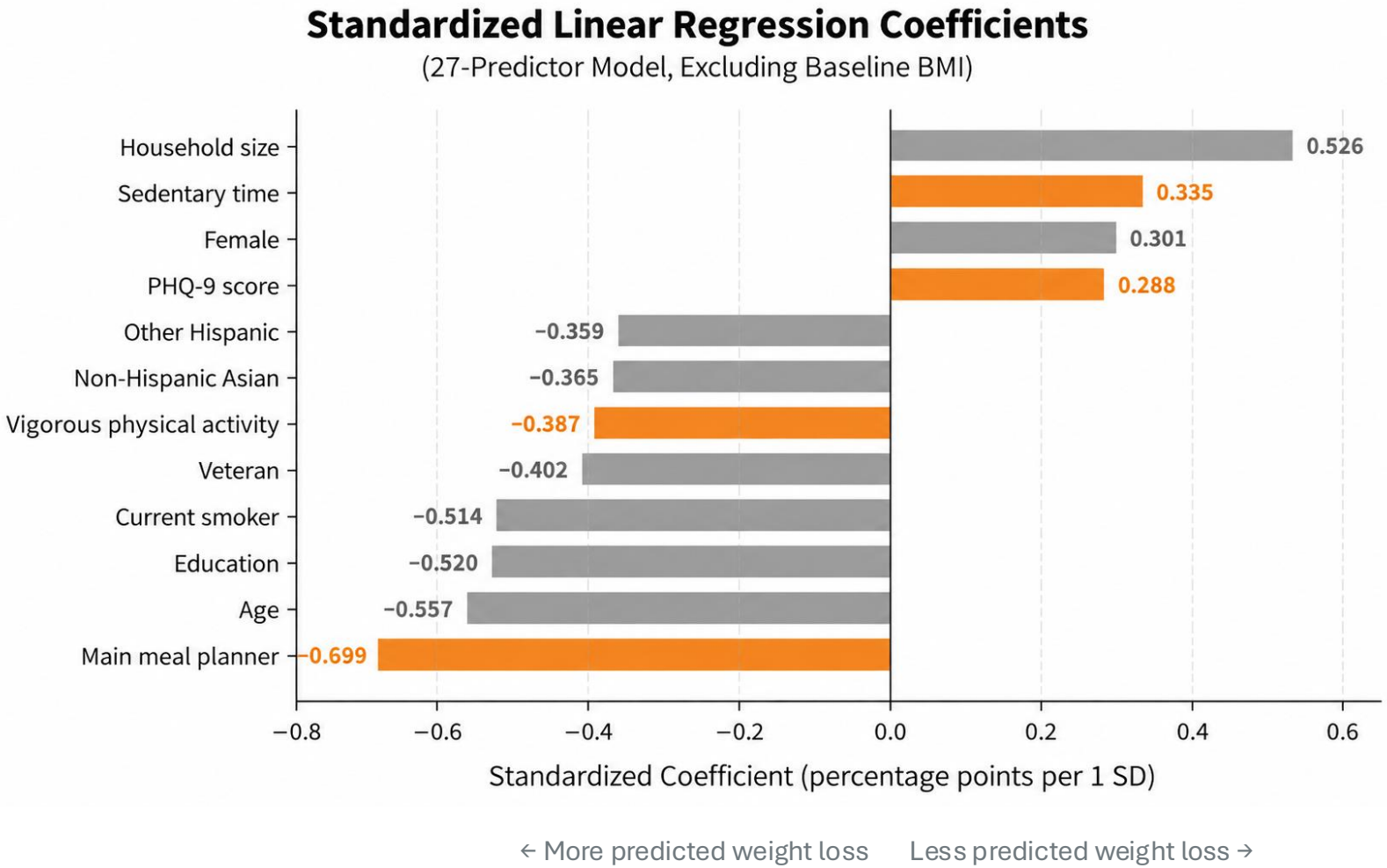
Model / predictors	Test R^2	CV R^2	Test RMSE
OLS: Covariates	0.068	0.116	8.765
OLS: Covariates + Behaviors	**0.085**	**0.121**	**8.687**
RF: Covariates	0.073	0.116	8.744
RF: Covariates + Behaviors	**0.081**	**0.122**	**8.707**

Added predictive value of behaviors

- **OLS:** Test ΔR^2 +0.017; 95% CI [−0.003, 0.037]
- **RF:** Test ΔR^2 +0.008; 95% CI [−0.008, 0.027]

Note. Both CIs include zero; OLS slightly better on test data, with negligible difference.

Models 3–4: Which Predictors Matter?



Note. **BMI excluded for scale: Coefficient = −3.172**

Key Findings

- **Starting BMI dominates:** OLS coefficient −3.172; RF importance +1.116 RMSE points.
- **OLS associations:** Meal planning and vigorous activity linked to more weight loss; sedentary time and PHQ-9 to less.
- **Binary interpretation:** Participants coded as main meal planners had approximately 1.57 percentage points lower predicted weight change than the comparison group, adjusting for the other predictors.
- **RF interpretation:** Moderate physical activity ranked third overall and highest among behavioral predictors in permutation importance. However, its contribution to predictive accuracy was small (RMSE increase = 0.037 when shuffled).
- **Prediction:** Behavioral additions yield modest, uncertain improvement; no clear forest advantage.